

UEI608: BIO-MEDICAL INSTRUMENTATION

L	T	P	Cr
3	0	2	4.0

Course Objectives: The objective of this course is to introduce student to basic biomedical engineering technology and introduce different biological signals, their acquisition, measurements and related constraints.

Introduction of Bio-medical Instrumentation, Sources of Bioelectric Potentials and Electrodes: Introduction to man-instrument system, components of the man-instrument system, Physiological system of the body, Problems encountered in measuring a living system. Resting and action potentials, Propagation of action potentials, bioelectric potentials, Bio potential electrodes, Biochemical transducers. Review of transducers

Cardiovascular System and Measurements: The heart and cardiovascular system, ECG, blood pressure and its measurement, respiration and pulse rate, characteristics and measurement of blood flow meter, cardiac output, plethysmography, pacemaker, defibrillators, heart sounds and its measurement,

Respiratory and Neuro-muscular System: The physiology of the respiratory system, test and instrument for the mechanics of breathing, the somatic nervous system, EEG, EMG and GSR, Disorders Related to the Respiratory and Neuromuscular Systems, Interconnection Between the Respiratory and Neuromuscular Systems.

Measurement and Recording of Non-invasive Diagnostic Instrumentation, Patient Care and Electrical Safety: Principle of ultrasonic measurement, ultrasonic, thermography, elements of intensive care monitoring, X-ray, CT – Scan and MRI, tonometer, dialysis, diathermy, Shock hazards from electrical equipment, Radiation Safety in Medical Imaging Future Trends in Non-invasive Diagnostic Instrumentation.

Laboratory work: Study the variance in pulse rate of subject in a batch, use Spiro meter on the subject, auditory system check-up using Audiometer, Measurement of Heart Rate using Stethoscope, Blood pressure using Sphygmomanometer, Pulse Rate and SpO₂ using Pulse Oximeter, Skin Conductance and Skin Potential using Galvanic Skin Response Module, Pulse Rate using Polyrite machine, Respiration Rate using Polyrite. Electromyogram test using EMG biofeedback Trainer.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

1. Distinguish and analyze the biomedical signal sources.
2. Analyze cardiovascular system and its measurement techniques.
3. Analyze the working principle and applications of respiratory and neuromuscular systems.
4. Evaluate the parameters non-invasive diagnostic instruments.
5. Study the electrical safety related to biomedical Instruments

Text Books:

1. *Cromwell, L. and Weibell, F.J. and Pfeiffer, E.A., Biomedical Instrumentation and Measurement, Dorling Kingsley (2006) 2nd ed.*
2. *Carr, J.J. and Brown, J.M., Introduction to Biomedical Equipment Technology, Prentice Hall (2000) 4th ed.*

Reference Books:

1. *Khandpur, R.S., Handbook of Biomedical Instrumentation, McGraw Hill (2003) 2nd ed.*
2. *Webster, J.G., Medical Instrumentation Application and Design, John Wiley (2007) 3rd ed.*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30